Power and energy

Energy makes things happen—from the smallest pulse of electricity to a blast of explosives. Energy is measured in joules. Power is the rate at which energy is converted from one form to another.

Measuring power

Power can be calculated by taking the amount of energy converted and dividing it by the time taken. The more energy that is converted in a set period or the quicker a specific amount of energy is converted, the greater the power. So an 1,800-watt electric heater can convert three times as much heat energy per second as a 600-watt model.

WHAT IS TORQUE?

Torque is a measure of the amount of twisting or turning force generated. It is most commonly used to describe an engine's "pulling power."

UNITS OF POWER

Power is measured in a number of ways and exhibited by many different things, including engines, appliances, and people.

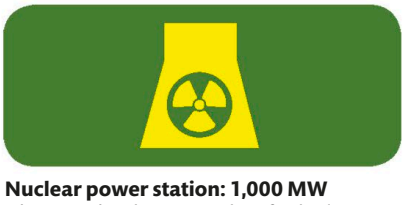
Watt (W)
A watt is equal to 1 joule of work done or 1 joule of energy converted per second. The rate at which a light bulb converts electrical energy into light is measured in watts. The higher the number of watts, the higher the power.

Kilowatt (kW) and megawatt (MW)
A kilowatt is 1,000 watts and is a useful measure of electricity used by large engines and appliances. A megawatt (MW) is equal to 1,000,000 watts. Only huge machines can generate power on this scale, including power plants, aircraft carriers, and supercolliders used for particle physics experiments.

Kilowatt-hour (kWh)
One kWh is equal to 1,000 watts used for 60 minutes, or 3.6 million joules. The amount of electrical energy used in a house is usually monitored and measured in kilowatt-hours (kWh).

Horsepower (hp)
The power of vehicle engines is often measured in horsepower (hp). One horsepower is equal to 746 watts. Brake horsepower (bhp) simply means that an engine's energy loss due to friction has been accounted for.

Power production and usage
How we think about and measure power depends on the object or task carried out. For some objects, "power" refers to how much power is produced, while for others, it indicates the amount of power used.



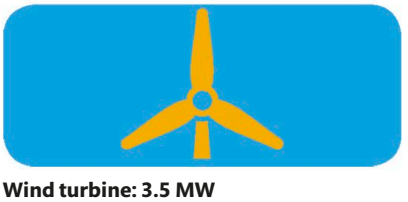
Nuclear power station: 1,000 MW
Like a wind turbine, a nuclear facility's power is often considered in terms of how much electricity it can generate when running at optimal capacity.



Microwave oven: 1,000 W
Microwave ovens are measured in terms of how much power they consume (for example, 1,000 W) and how much energy they consume in a year (typically 62 kWh).



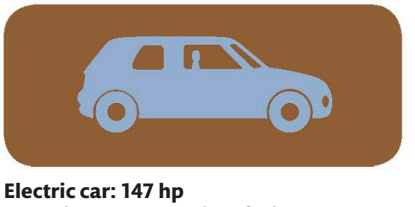
Gas-engine supercar: 1,479 hp
A car engine's peak horsepower refers to its maximum power output. Some supercars, such as the Bugatti Chiron, can reach up to 1,479 hp.



Wind turbine: 3.5 MW
A typical offshore wind turbine can produce up to 3.5 MW of electricity each year—enough to supply power to about 1,000 households.



LED TV: 60 W
Although an LED TV has a far lower power rating (typically 60 W) than a microwave, it is in use far more, so its annual energy consumption (around 54 kWh) is similar.



Electric car: 147 hp
Most electric cars produce far less power than gas engines, but their electric motors generate more torque at a standstill and at low speeds.